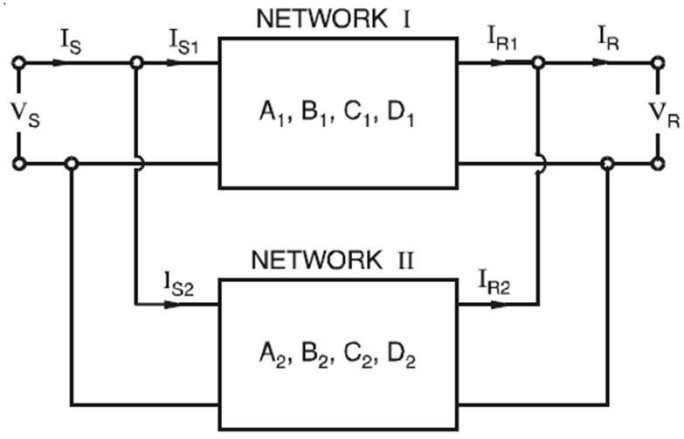


Fig. 6.28. *Two Networks in Parallel*

$$\mathbf{A} = \frac{\mathbf{A_1B_2 + A_2B_1}}{\mathbf{B_1 + B_2}}$$

$$\mathbf{B} = \frac{\mathbf{B_1B_2}}{\mathbf{B_1 + B_2}}$$

$$\mathbf{C} = \mathbf{C_1 + C_2} - \frac{\mathbf{(A_1 - A_2)(D_1 - D_2)}}{\mathbf{B_1 + B_2}}$$

$$\mathbf{D} = \frac{\mathbf{B_1D_2 + B_2D_1}}{\mathbf{B_1 + B_2}}$$

$$\text{Denominator} = B_1 + B_2$$

$$A = \frac{A_1B_2 + A_2B_1}{B_1 + B_2}$$

B has only B in its equation

$$D = \frac{D_1B_2 + D_2B_1}{B_1 + B_2}$$

Calculate C using AD-BC = 1

Replace A by D, and the equation remains identical